

Sai Meghana Barla

980-535-5319 | sbarla1@ufl.edu | [linkedin/saim meghana-barla03](https://www.linkedin.com/saim meghana-barla03) | github.com/meghana21-arch

Software engineer and MS CS student building production-grade backend systems, AI developer tools, and distributed workflow platforms. Scaled TrueCaptcha.org from ~1K to 1M+ daily API requests with 99.99% uptime and reduced latency from ~7s to ~1s. Built AgentOps Runtime, a durable execution layer for AI agent loops with crash recovery, retries, state persistence, and observability.

EDUCATION

University of Florida - MS in Computer Science, GPA: 4.0 / 4.0

Jan 2026 – Dec 2027

EXPERIENCE

Research Assistant | University of Florida

Jan 2026 – Present

- Scaled a distributed workflow orchestration system to support 1K+ users and daily HPC job execution.
- Built an ML-driven scheduling system to reduce job wait times and improve cluster utilization for distributed workloads.
- Developed a full-stack React workflow UI for distributed systems, reducing setup time by ~90%.

Co-Founder & Founding Software Engineer | TrueCaptcha.org

Aug 2020 – Jan 2026

- Built and scaled a production AI SaaS platform generating ~\$20K revenue and serving 1M+ daily API requests, with a 2.8M request peak.
- Designed distributed APIs and backend systems for high-concurrency OCR workloads, maintaining 99.99% uptime while supporting real-time request processing at production scale.
- Reduced end-end API latency from ~7s to ~1s by optimizing Redis caching, async worker queues, horizontal scaling, and distributed processing pipelines.
- Built developer-facing APIs, usage dashboards, and monitoring workflows used by 300+ daily users, improving integration workflows and reducing onboarding friction.

Software Engineer | Digit Insurance

Jun 2024 – Nov 2025

- Re-architected 80 legacy batch workflows into modular Java Spring Boot services, reducing production debugging cycles from days to same-day root-cause isolation.
- Designed 30+ PostgreSQL-backed REST APIs for policy, premium, and customer data workflows, reducing internal data retrieval delays from hours to near real-time responses.
- Optimized batch pipelines processing millions of insurance records using query tuning, indexing, and access-pattern improvements, reducing execution time by 30%.
- Improved backend observability with structured logs and debugging workflows, reducing mean time to root-cause analysis from hours to under 30 minutes.

Software Engineer Intern | Keysoft IT Private Limited

Aug 2022 – Mar 2023

- Streamlined operations using Strapi CMS integrated with Next.js, optimizing product management and user order processes, resulting in a 41% increase in weekly orders.
- Implemented backend workflows and integrations to improve product management, order tracking, and user operations.

PROJECTS

AgentOps Runtime – Durable Execution Layer for AI Agent Loops | TypeScript, Go, PostgreSQL, Docker, AWS, Claude API, Redis

- Built a durable execution runtime for Claude-powered agent loops, handling 1,000+ simulated multi-step tasks with planning, tool execution, retries, approvals, and crash recovery.
- Designed a PostgreSQL-backed state machine with transactional event persistence, preventing state corruption during 100+ simulated worker crashes and workflow restarts.
- Implemented Redis retry queues, idempotency keys, and exponential backoff, reducing duplicate tool executions across failed and retried agent runs.
- Optimized agent execution tracking with structured traces and run-level metrics, reducing debugging time from hours to under 30 minutes.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, Bash | **Backend:** Spring Boot, FastAPI, Node.js, Express.js, REST APIs, GraphQL, Microservices | **AI Infra:** LLM APIs, RAG, AI Agents, Evaluation Systems, LLM Observability, OCR | **Databases:** PostgreSQL, MySQL, Redis, DynamoDB | **Infra:** Docker, Kubernetes | **Cloud & Tools:** AWS EC2, S3, Lambda, ECS, ECR, Git, Linux, CI/CD